

6.1120 Fall 2025: Recitation 4

Phase 2 Overview

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Announcements

1. Phase 1

- Reminder that Phase 1 is due **10pm** today (Monday, September 22).
- Please submit to Gradescope *as soon as possible* to make sure there are no build issues.

2. Phase 2

- Phase 2 has officially released (as of 12pm).
- Phase 2 will be due at **10pm** on **Friday, October 3**.
- *Please start early!*

3. Miniquiz

- There is a miniquiz due tomorrow at 11:59pm.
- Forgot to release it on Friday, so I extended the time.

```
// cat arith.mit  
sum = 1 + 1;  
print("1 + 1 = " + sum);
```

```
// ./run.sh interpret arith.mit  
1 + 1 = 2
```

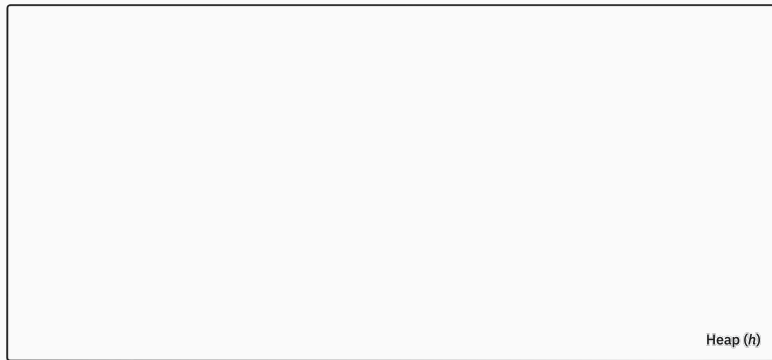
All the following deliverables are due, **October 3 at 10pm**.

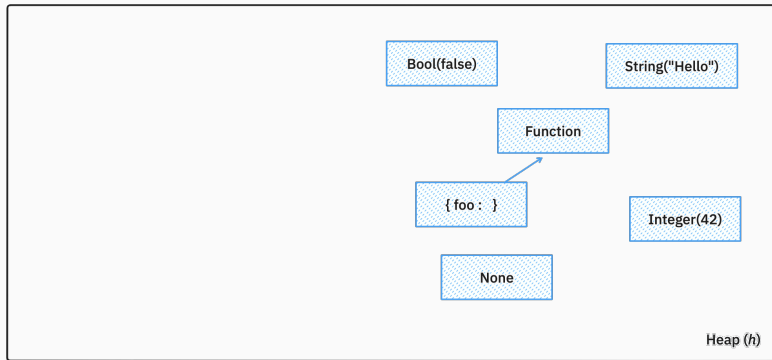
1. **Implementation**

- Must conform to the semantics described in the language specifications.
- Must be able to build your code using `./build.sh` (again submit to Gradescope early, esp. if you changed the build system).

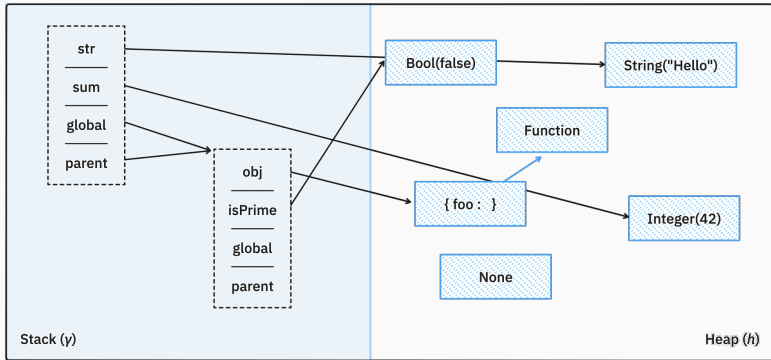
2. **Short report** (up to 3 paragraphs).

3. **Ten additional test cases.**





Semantics



The style of semantics covered in 6.1120 is sometimes called *operational semantics*. This style is very popular for constructing proofs from inference rules which talk about a program's execution.

In particular, MITScript defines its semantics using **small-step operational semantics**.

Semantics (Statements)

$$\text{VAR} \frac{(\gamma; a_1, h, e) \rightarrow (h', v) \quad \text{lookup-write}(a_1, h', x) = a_2 \quad \sigma = h'(a_2) \quad a_3 \notin \text{dom}(h') \quad \sigma' = \sigma[x : a_3]}{(\gamma; a_1, h, x = e) \rightarrow h'[a_3 : v][a_2 : \sigma']}$$

$$\text{IFFALSE} \frac{(\gamma, h, e) \rightarrow (h', \text{Bool}(\text{false})) \quad (\gamma, h', s_2) \rightarrow \eta}{(\gamma, h, \text{if } e \text{ } s_1 \text{ else } s_2) \rightarrow \eta}$$

$$\text{IFTRUE} \frac{(\gamma, h, e) \rightarrow (h', \text{Bool}(\text{true})) \quad (\gamma, h', s_1) \rightarrow \eta}{(\gamma, h, \text{if } e \text{ } s_1 \text{ else } s_2) \rightarrow \eta}$$

$$\text{S} \frac{(\gamma, h, s_1) \rightarrow h' \quad (\gamma, h', s_2) \rightarrow \eta}{(\gamma, h, s_1; s_2) \rightarrow \eta}$$

$$\text{S-R} \frac{(\gamma, h, s_1) \rightarrow \text{Return}(h', v)}{(\gamma, h, s_1; s_2) \rightarrow \text{Return}(h', v)}$$

Semantics (Expressions)

$$\text{INTCONST} \frac{}{(\gamma, h, n) \rightarrow (h, \text{Integer}(n))}$$

$$\text{ARITHOP} \frac{(\gamma, h, e_1) \rightarrow (h', \text{Integer}(n_1)) \quad (\gamma, h', e_3) \rightarrow (h'', \text{Integer}(n_2))}{(\gamma, h, e_1 \text{ op } e_2) \rightarrow (h'', \text{Integer}(n_1 \text{ op } n_2))}$$

$$\text{VARREAD} \frac{\text{lookup-read}(a_1, h, x) = a_2 \quad h(a_2) = v}{(\gamma; a_1, h, x) \rightarrow (h, v)}$$

$$\text{FUNCTION} \frac{m = \{\text{ctx} : a, \text{formal} : x, \text{body} : s\} \quad a_f \notin \text{dom}(h) \quad h' = h[a_f : m]}{(\gamma; a, h, \text{fun } x \text{ } s) \rightarrow (h', \text{Function}(a_f))}$$

Thank you!